

NIEDERSTETTEN, Germany

WMO: 107430

Lat: 49.3894N

Lon: 9.9667E

Elev: 459

StdP: 95.93

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.9	-8.3	-13.1	1.3	-9.9	-10.5	1.6	-6.9	15.2	6.3	13.3	5.9	3.4	50	0.555

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	30.3	19.2	28.2	18.4	26.2	17.8	20.1	27.9	19.3	26.5	18.4	25.0	3.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	13.3	22.6	16.8	12.7	22.0	16.0	12.0	21.0	59.7	28.0	56.7	26.4	54.0	24.9	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.3	7.8	DB	-13.2	33.5	2.9	1.7	-15.3	34.7	-17.0	35.7	-18.6	36.7	-20.7	38.0	(4)
(5)			WB	-13.5	22.0	2.9	1.1	-15.6	22.7	-17.3	23.4	-18.9	24.0	-21.1	24.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	-0.1	1.1	4.9	9.1	13.2	16.6	18.5	18.3	14.0	9.7	4.4	1.3	(6)
(7)		DBStd	7.71	4.64	4.61	4.13	4.25	3.87	3.87	3.64	3.64	3.48	3.89	3.90	4.26	(7)
(8)		HDD10.0	1308	312	248	164	68	13	1	0	0	5	54	171	272	(8)
(9)		HDD18.3	3432	571	481	417	279	165	78	44	46	137	269	417	529	(9)
(10)		CDD10.0	1050	0	1	6	39	113	199	263	258	124	44	3	0	(10)
(11)		CDD18.3	134	0	0	0	0	6	27	49	46	6	1	0	0	(11)
(12)		CDH23.3	1272	0	0	0	5	62	261	463	420	59	2	0	0	(12)
(13)		CDH26.7	373	0	0	0	1	8	68	149	138	9	0	0	0	(13)
(14)	Wind	WSAvg	3.5	4.3	4.1	4.1	3.3	3.1	2.9	3.1	3.0	3.1	3.4	3.7	4.1	(14)
(15)	Precipitation	PrecAvg	769	61	58	60	47	67	69	77	63	61	67	67	72	(15)
(16)		PrecMax	1015	129	159	156	132	146	162	138	139	131	207	121	203	(16)
(17)		PrecMin	517	10	16	8	2	16	26	16	12	20	18	2	10	(17)
(18)		PrecStd	113	30	35	38	28	34	31	34	33	29	37	30	41	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	14.1	18.1	23.9	27.8	31.4	33.1	33.8	28.0	22.8	15.7	11.2	(19)
(20)			MCWB	8.3	8.7	10.6	14.0	18.0	20.2	19.8	19.3	17.6	15.6	12.1	8.3	(20)
(21)		2%	DB	9.6	10.9	15.8	21.1	25.0	28.6	30.3	30.2	25.0	19.2	12.9	9.8	(21)
(22)			MCWB	7.5	7.1	9.5	12.5	16.9	19.0	19.1	19.0	16.7	14.5	10.3	8.1	(22)
(23)		5%	DB	7.8	8.9	13.1	18.9	22.6	26.2	28.1	27.8	22.5	17.0	11.2	8.3	(23)
(24)			MCWB	6.1	6.4	8.2	11.7	15.1	18.2	18.3	18.2	15.8	13.4	9.1	7.0	(24)
(25)		10%	DB	6.1	7.1	10.9	16.4	20.4	24.1	25.9	25.3	20.1	15.1	9.9	7.0	(25)
(26)			MCWB	4.9	5.2	7.6	10.5	14.0	17.3	17.6	17.4	14.8	12.4	8.2	5.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	9.2	11.6	14.7	19.1	21.7	21.0	20.9	18.6	16.5	12.5	9.6	(27)
(28)			MCDWB	10.6	12.6	16.9	21.7	26.0	28.6	29.7	29.2	25.1	20.9	14.6	10.5	(28)
(29)		2%	WB	7.6	7.8	10.1	13.2	17.5	20.0	19.9	19.8	17.3	14.9	10.7	8.3	(29)
(30)			MCDWB	9.3	10.1	13.6	19.8	23.7	26.7	28.2	27.7	23.2	18.3	12.5	9.4	(30)
(31)		5%	WB	6.1	6.6	9.0	12.1	16.0	18.7	19.1	19.0	16.4	13.8	9.5	7.0	(31)
(32)			MCDWB	7.4	8.5	12.2	17.7	21.4	25.1	26.3	26.3	21.4	16.5	10.9	8.2	(32)
(33)		10%	WB	4.8	5.3	7.8	10.9	14.6	17.6	18.2	18.0	15.3	12.7	8.3	5.8	(33)
(34)			MCDWB	5.9	6.9	10.6	15.9	19.5	23.5	24.6	24.4	19.4	14.9	9.6	6.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.8	6.1	7.5	9.4	9.7	10.1	10.1	10.0	8.8	7.3	5.1	4.4	(35)
(36)			MCDBR	6.5	8.7	11.3	13.2	13.1	13.6	14.1	13.9	12.6	10.1	7.6	5.9	(36)
(37)			MCWBR	5.1	6.0	6.7	7.0	6.9	6.7	5.8	5.8	6.1	6.0	5.3	4.7	(37)
(38)		5% WB	MCDBR	6.0	7.6	9.6	12.0	12.0	12.6	12.0	12.3	10.8	9.0	6.9	5.4	(38)
(39)			MCWBR	5.0	5.6	6.1	6.6	6.7	6.5	5.4	5.6	5.7	5.7	5.1	4.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.299	0.323	0.356	0.387	0.395	0.398	0.403	0.398	0.369	0.347	0.322	0.298	(40)	
(41)		taud	2.438	2.388	2.329	2.270	2.271	2.290	2.269	2.309	2.370	2.429	2.454	2.436	(41)	
(42)		Ebn at Noon	760	822	850	858	864	862	850	837	826	782	721	707	(42)	
(43)		Edh at Noon	63	83	104	123	128	127	127	117	99	79	62	55	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.86	1.65	2.79	4.28	5.06	5.64	5.33	4.62	3.38	2.00	0.98	0.68	(44)	
(45)		RadStd	0.14	0.27	0.38	0.56	0.56	0.46	0.52	0.44	0.42	0.22	0.11	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.49	N/A	N/A	N/A	N/A	N/A	N/A	-150	+73	N/A
(47) Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.33	N/A	N/A	N/A	N/A

Nomenclature: See separate page